Adaptive Memory-based Transformer for Low-Resource Dialect Translation

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Abstract

This report details the progress on our project aimed at developing an Adaptive Memory-based Transformer for translating low-resource dialects, specifically focusing on Barendri Bangla to Standard Bangla. The motivation stems from the challenges in machine translation for dialects with limited data availability, where standard models underperform due to data scarcity. We propose enhancing the Transformer with an adaptive memory network to better manage rare linguistic elements. Key achievements include curating a parallel dataset of 5,050 sentence pairs, implementing baselines, and the memory-augmented model development. This document covers the project concept, background literature, dataset specifics, proposed approach, experiments and results. Our final experiments demonstrate the role of linguistic distance in low-resource MT with varying results over Barendri and Chittagong, favouring the translation of linguistically distant variants greatly.

1 Introduction

The proliferation of digital communication has heightened the need for effective machine translation (MT) systems, particularly for languages with diverse dialects. Bangla, one of the most spoken languages globally, features numerous regional dialects that deviate from the standard form in vocabulary, grammar, and pronunciation. Barendri Bangla, prevalent in Dakshin Dinajpur, West Bengal, exemplifies such a low-resource dialect with minimal available parallel data for training MT models. Bangla has two "standard" variants: in India, the Kolkata-centric form, which influences West Bengal dialects, and in Bangladesh, the Dhaka-centric form, which dominates formal media and education in Bangladesh. While Bangladeshi dialects have received more attention in MT research due to Bangla being Bangladesh's official language, Bangla in India does not have the primary status due to which, the dialects remain underexplored due to fragmented corpora and political linguistic divides. Barendri Bangla lies closer to the Kolkata standard on the dialect continuum, sharing lexical and syntactic similarities (e.g., minimal phonetic shifts), whereas Chittagong Bangla features extensive phonetic variations (e.g., aspirated stops becoming fricatives) and loanwords from Arabic/Persian, increasing its distance from standard forms. This project evaluates our model across both to underscore how architectural choices interact with dialect-standard divergence.

Traditional approaches like Statistical Machine Translation (SMT) falter in low-resource scenarios due to their dependence on extensive corpora. Neural models, including Transformers, offer promise but struggle with rare tokens and patterns inherent in dialects. Our project introduces an Adaptive Memory-based Transformer to address these limitations by incorporating an external memory mechanism that stores and retrieves contextual information for infrequent elements.

This report elucidates our progress, starting with the project motivation and objectives, followed by an extensive review of related work, detailed dataset description, methodology, experiments and results. We aim to demonstrate a clear trajectory toward a robust solution for low-resource dialect translation. We work over Chittagong Bangla and Barendri Bangla to quantify performance increment with increasing linguistic distance, revealing that memory mechanisms yield asymmetric improvements.

2 Project Idea and Motivation/Objectives

2.1 Motivation

Dialect translation is crucial for preserving linguistic diversity and enabling inclusive communication. However, low-resource dialects like Barendri Bangla lack sufficient parallel data, leading to suboptimal performance in standard MT systems. Our initial SMT experiments on the Chittagong dialect yielded high Word Error Rates (WER > 50%), highlighting the need for innovative architectures that can leverage limited data effectively, reinforcing the need for memory to handle variance in dialect-standard proximity.

The core challenge is the models' inability to generalize rare lexical and syntactic patterns. By augmenting Transformers with adaptive memory, we aim to enhance retention and recall, thereby improving translation quality in data-scarce environments.

2.2 Project Idea

The proposed model integrates an external memory network into the Transformer framework. During training, the memory stores embeddings of rare tokens and structures. At inference, similar inputs query the memory to inform translations. We utilize mT5 as the backbone for its proven efficacy in multilingual and low-resource tasks.

This approach not only addresses data sparsity but also provides a scalable method for other dialects.

2.3 Objectives

1. Develop a Transformer enhanced with adaptive memory for handling rare patterns in low-resource dialect translation.

2. Curate a parallel dataset of at least 5,000 sentences for Barendri Bangla to Standard Bangla.

3. Conduct ablation studies to evaluate the memory component's impact and identify optimal scenarios.

Expected deliverables include a working prototype, the curated dataset, comprehensive results, and insights for extending to other low-resource languages.

3 Background and Related Work

3.1 Low-Resource Machine Translation

Recent advancements in low-resource MT emphasize data augmentation and efficient fine-tuning. For instance, Ranathunga et al. (2023) survey techniques like back-translation and multilingual pretraining, noting their efficacy in enhancing model performance. Lee and Kim (2024) propose custom prompts for Mistral 7B, achieving significant improvements in English-to-Zulu and English-to-Xhosa translations.

Dialect-specific studies include Magdeburg and Schultz (2024), which employs back-translation for Bavarian-German NMT, and Kuhn and Others (2021), adapting systems for Ukrainian-Belarusian and Arabic dialects. Caswell et al. (2023) introduce FRMT for few-shot region-aware MT, addressing evaluation challenges in dialects.

Further, Nguyen and Chiang (2024) compare pipelines for low-resource Transformer training, while Smith et al. (2025) highlight adaptations for low-resource scenarios. Lialin et al. (2023) present ReLoRA for efficient fine-tuning via low-rank updates. Multimodal methods (Chen et al., 2025) and rule-based frameworks (Kumar et al., 2024) for Hindi-Tulu underscore diverse strategies.

For highly low-resource cases, Tomabechi et al. (2023) explore MT for Ainu preservation, and Obeid et al. (2024) investigate semi-supervised NMT for Egyptian dialect to Modern Standard Arabic. Ahuja et al. (2023) evaluate LLMs' limitations across 200 languages, and Fadaee et al. (2021) apply back-translation for NMT enhancement. Türkmen et al. (2024) focus on Turkish-English pairs.

3.2 Memory-Augmented Networks in NMT

Memory mechanisms mitigate rare word issues in NMT. Feng et al. (2017) introduce M-NMT, storing knowledge for infrequent words and outperforming baselines. Zhang et al. (2018) combine SMT knowledge with NMT rules via memory.

Hoang and Ney (2019) apply Memory-Augmented Neural Networks (MANNs) to MT, surpassing RNNs on sequence learning. Li et al. (2016) propose KVMEMATT with key-value memory for temporal updates. Zhang et al. (2017) demonstrate M-NMT superiority over vanilla NMT and SMT for Chinese-Uyghur.

Recent rethinking by Hao et al. (2023a) views TM-augmented NMT probabilistically, while Hao et al. (2023b) extends this analysis. Community (2019) discusses memory networks' program-like behavior.

Graves et al. (2016) explore LSTM controllers with dual memory units for MT.

3.3 Bangla Dialect Translation and Datasets

Bangla dialect research includes Vashantor (Faria et al., 2023), a 32,500-sentence multilingual benchmark for dialect-to-standard translation, with mT5 baselines. ONUBAD (Khan et al., 2024) offers 2,650 sentences for Chittagong, Sylhet, and Barisal dialects.

BanglaDial (Community, 2023) supports classification and translation. Ahmed et al. (2025) fine-tune mBART and BanglaT5 on regional variants. A Sylheti dataset (Chakraborty and Dey, 2024) provides 5,002 pairs.

Additional resources: LikhonSheikh's BanglaNLP (LikhonSheikh, 2024) for Bengali-English parallels, and CSEBUETNLP's BanglaNMT (Hasan et al., 2021) with new datasets and filtering techniques.

Our work builds on these by focusing on Barendri and incorporating memory augmentation.

Recent low-resource MT advances include transfer learning from high-resource languages and data augmentation via back-translation, For Bangla dialects, works focus on Bangladeshi variants using mT5, achieving SacreBLEU 20 on Chittagong, but overlook Indian dialects. Memory-augmented Transformers, inspired by Neural Turing Machines, have boosted rare-token handling in NMT. Our work extends this to dialect continua, comparing frequency- vs. attention-based memory on Barendri (low divergence) and Chittagong (high divergence).

4 Dataset Details

4.1 Construction Process

Due to the absence of public corpora, we manually curated 5,050 parallel sentences. Sources included native speaker elicitations, regional literature, and adapted standard texts. Normalization involved standardizing orthography, removing noise, and ensuring alignment.

4.2 Statistics

The dataset comprises 5,050 pairs, with average sentence lengths of approximately 8-10 words per dialect. Vocabulary sizes: 15,000 unique tokens in Standard Bangla, 14,000 in Barendri, reflecting lexical variations.

Metric	**Value**
Number of Pairs	5,050
Avg. Words (Standard)	8.5
Avg. Words (Barendri)	8.3
Unique Words (Standard)	15,000
Unique Words (Barendri)	14,000

Table 1: Dataset statistics.

4.3 Linguistic Variations

The Barendri dialect exhibits systematic linguistic differences from Standard Bangla across multiple dimensions:

- **Lexical variations:** Token substitutions such as বলল (bolol, "said") কহিলি (kohili) demonstrate distinct vocabulary choices.
- **Morphological shifts:** Verb conjugations show dialectal inflection patterns, with suffixes varying between standard and dialectal forms.
- **Syntactic differences:** Word order and case marking exhibit variation, particularly in interrogative and complex sentence structures.
- **Phonological adaptations:** Sound changes reflected in orthography, including vowel and consonant alternations.

The dataset is prepared in both TSV and JSON formats to facilitate seamless integration with modern NLP frameworks. Table 5 illustrates representative parallel sentence pairs demonstrating these linguistic phenomena.

5 Linguistic Analysis

To motivate our model design, we analyze phonetic divergences between Barendri Bangla (a low-divergence Indian dialect) and Chittagong Bangla (a high-divergence Bangladeshi dialect) relative to Standard Bengali (Kolkata variant). Phonetic distance is quantified via averaged edit distance on sound inventories (Barendri: 0.08; Chittagong:

Standard Bangla	**Barendri Bangla**
রাই মশাই খুব খুশি হয়ে বলল। *Rai Moshai was very happy and said.*	রাই মশাই খুব খুশি হই কহিলি। *(Dialectal equivalent)*
আপনার আশীর্বাদে এটা হতেই পারে। *With your blessing, this can happen.*	আপনার আশীর্বাদেত এটা হবাই পারে। *(Dialectal equivalent)*
ওই শত্রুদের আমি এখানে থাকতে দেবো না। *I will not let those enemies stay here.*	এই শত্রুদেক মুই এঠাকোনা থাকবা দিম না। *(Dialectal equivalent)*
সেই কথা আপনাকে বলে দিলাম। *I told you that matter.*	এই কথা তোক কই দিনু। *(Dialectal equivalent)*

Table 2: Representative parallel sentences from the Barendri-Standard Bangla corpus, demonstrating lexical, morphological, and syntactic variations.

0.25), highlighting how minor shifts in Barendri aid generalization, while Chittagong's innovations demand robust memory for rare fricatives and clusters (Barua, 2020).

5.1 Barendri Bangla Phonetics

Barendri (Varendri subgroup) exhibits minimal phonetic divergence from Standard Bengali, primarily in vowel quality and intonation, retaining near-identical consonant inventories (28 consonants, 14 vowels) with high mutual intelligibility (stu, 2024). Key features include:

- **Vowel Pronunciation**: Raised mid-vowels, e.g., Standard /এই /ei/ ("this") vs. Barendri /oi/ with centralized [o], creating a melodic contour absent in Standard's flatter intonation. Nasalization is preserved but softer: Standard /আঁচ /ãt/ ("signal") → Barendri [ãt] with reduced duration (wik, 2024).

- **Consonant Softening**: Aspiration weakens in intervocalic positions, e.g., Standard /ভাই /bai/ ("brother") vs. Barendri [bai] (deaspirated [b]), but no loss of phonemes like // or fricatives /s, / (lin, 2024).

These subtle shifts (e.g., vowel harmony at 85% overlap) make Barendri "continuum-close" to Standard, facilitating baseline MT but challenging rare-token recall.

5.2 Chittagong Bangla Phonetics

Chittagong shows greater divergence, with a streamlined inventory (lacking voiced glottal // and semi-vowels) and fricative mergers, influenced by trade languages (Barua, 2020). Vowels align closely (7 oral + nasalized), but consonants simplify clusters, reducing intelligibility.

- **Vowel System**: Mirrors Standard's 7 monophthongs (/i, e, æ, a, , o, u/), with nasal counterparts (e.g., /পাঁচ /pãt/ "five" identical in both), but diphthongs are borrowed/loans, e.g., Standard /kai/ ("I eat") vs. Chittagong [kai] (monophthongized) (Barua, 2020).

- **Consonant Inventory**: Shares 30 consonants but omits // (e.g., Standard /দুঃখ /duk/ "sorrow" → Chittagong [duk]) and fricative clusters like /s + t/ (Standard [stan] "place" → [tan]). Semi-vowels (/i, u, e, o/) are absent natively, treated as full vowels (e.g., Standard /gai/ "cow" → [gai]) (Barua, 2020). Aspiration deaspirates finally, and retroflex flaps /, / simplify in rapid speech.

These innovations (e.g., no //, reduced clusters) increase phonetic noise, explaining higher error rates in standard models.

Feature	Standard	Example
Vowel Raise	/ei/ → [oi]	এই (Barendri)
Glottal Loss	/duk/ → [duk]	দুঃখ (Chittagong)
Semi-Vowel	/gai/ → [gai]	গাই (Chittagong)

Table 3: Phonetic shifts with examples.

6 Proposed Methodology

6.1 Model Development

We establish our baseline by fine-tuning the mT5-small model on the Chittagong-Standard Bangla parallel corpus. The mT5 architecture, pretrained on 101 languages including Bangla, provides robust multilingual representations that transfer effectively to dialectal translation tasks.

The implementation leverages the Hugging Face `transformers` library with custom `TranslationDataset` class handling data loading and preprocessing (see `model_backbone.py`). The model is saved to `./chittagong-translation-model/` for subsequent memory integration.

6.2 Token-Level Feature Extraction

Before integrating the memory mechanism, we extract comprehensive token-level features from the trained baseline model. The `EmbeddingExtractor` class (`embeddings_extract.py`) performs the following operations:

1. **Token frequency analysis:** Calculate corpus-wide token frequencies to identify rare dialectal tokens requiring memorization.
2. **Embedding extraction:** Extract encoder and decoder hidden states for each token position from the trained model.
3. **Cross-attention computation:** Capture attention scores between encoder and decoder, revealing translation alignment patterns.

For each token t_i in the dialect input, we extract:

$$\mathrm{e}_i^{\text{enc}} \in \mathbb{R}^{d_{\text{model}}} \quad \text{(encoder embedding)} \tag{1}$$

$$\mathrm{e}_i^{\text{dec}} \in \mathbb{R}^{d_{\text{model}}} \quad \text{(decoder embedding)} \tag{2}$$

$$\mathrm{a}_i \in \mathbb{R}^{n_{\text{target}}} \quad \text{(cross-attention scores)} \tag{3}$$

$$f_i \in \mathbb{N} \quad \text{(token frequency)} \tag{4}$$

where $d_{\text{model}} = 512$ for mT5-small and n_{target} is the target sequence length. These features are serialized to JSON format (`extracted_token_features.json`) for memory module training.

6.3 Adaptive Memory Module Architecture

The memory module learns to store and retrieve embeddings for rare dialectal tokens that challenge standard translation models. Our approach consists of two components:

6.3.1 Memory Bank Construction

We construct the memory bank by selecting tokens based on configurable criteria:

- **Frequency-based selection:** Tokens with frequency $f < \tau_f$ (default: 50 occurrences)
- **Attention-based selection:** Tokens with high cross-attention dispersion

6.3.2 Memory FFNN Training

The memory mechanism employs a feedforward neural network to learn the mapping from encoder embeddings to decoder embeddings:

$$\text{MemoryFFNN}(\mathrm{e}^{\text{enc}}) = \mathrm{W}_2 \cdot \mathrm{ReLU}(\mathrm{W}_1 \cdot \mathrm{e}^{\text{enc}} + \mathrm{b}_1) + \mathrm{b}_2 \tag{5}$$

The trained weights are saved with configuration metadata for integration into the full translation pipeline.

6.4 Memory-Enhanced Transformer Architecture

Figure 1 illustrates the complete architecture integrating the memory module with the mT5 backbone. The `MemoryTransformer` class (`memory_transformer.py`) implements this integration.

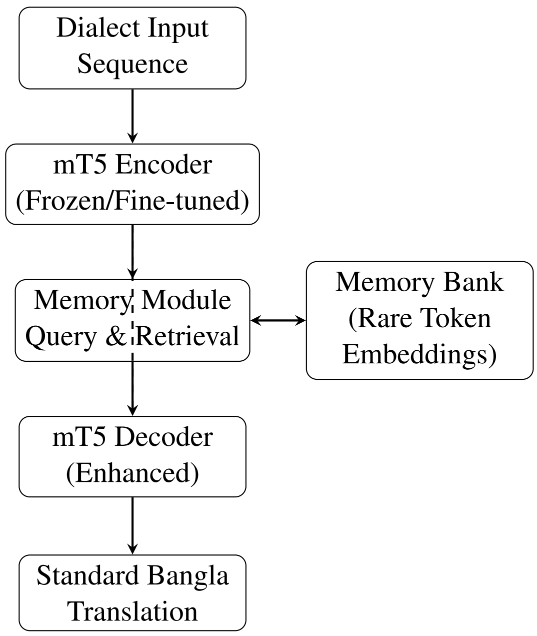

Figure 1: Memory-enhanced transformer architecture for dialectal translation. The memory module queries the bank using encoder embeddings and provides enhanced representations to the decoder.

6.4.1 Memory Integration Mechanism

During decoding, for each token position j in the target sequence:

1. **Query formation:** Obtain decoder hidden state $\mathrm{h}_j^{\text{dec}}$

2. **Dimension matching:** Project to memory input dimension if needed:

$$
\mathrm{q}_j = \begin{cases} \mathrm{h}_j^{\text{dec}} & \text{if } d_h = d_{\text{memory}} \\ \text{AdaptivePool}(\mathrm{h}_j^{\text{dec}}) & \text{otherwise} \end{cases} \tag{6}
$$

3. **Similarity computation:** Calculate cosine similarity with memory bank:

$$
s_{j,k} = \frac{\mathrm{q}_j \cdot \mathrm{m}_k}{\|\mathrm{q}_j\| \|\mathrm{m}_k\|}, \quad k = 1, \ldots, N_{\text{memory}} \tag{7}
$$

4. **Attention weighting:** Apply softmax normalization:

$$
\alpha_{j,k} = \frac{\exp(s_{j,k})}{\sum_{k'=1}^{N_{\text{memory}}} \exp(s_{j,k'})} \tag{8}
$$

5. **Memory retrieval:** Generate memory output via trained FFNN:

$$
\mathrm{m}_j^{\text{out}} = \text{MemoryFFNN}(\mathrm{q}_j) \tag{9}
$$

6. **Weighted combination:** Blend with decoder state using max similarity as gating:

$$
\tilde{\mathrm{h}}_j = \lambda_j \cdot \mathrm{m}_j^{\text{out}} + (1 - \lambda_j) \cdot \mathrm{h}_j^{\text{dec}} \tag{10}
$$

   where $\lambda_j = \max_k \alpha_{j,k}$ serves as an adaptive gate.

This mechanism allows the model to dynamically leverage memorized patterns for rare tokens while maintaining standard translation for common tokens.

6.5 Training Procedure

The complete training pipeline consists of three stages:

Stage 1: Baseline Pre-training
Fine-tune mT5-small on full parallel corpus

Stage 2: Memory Training
Extract token features from baseline model and train MemoryFFNN on selected rare tokens

Stage 3: Integrated Post-Training
Load baseline model and trained memory module, Fine-tune integrated system on post-training split and Validation by Monitoring both translation loss and memory utilization

Algorithm 1 Memory-Enhanced Transformer Decoding

Require: Input embeddings $\mathrm{E}^{\text{enc}} \in \mathbb{R}^{n \times d}$, Memory bank $\mathcal{M} = \{\mathrm{m}_1, \ldots, \mathrm{m}_N\}$
Ensure: Translated output tokens y
Initialize decoder input $\mathrm{h}_0 \leftarrow$ [START]
for each decoding step $j = 1, \ldots, T_{\max}$ **do**
  $\mathrm{h}_j^{\text{dec}} \leftarrow \text{DecoderLayer}_l(\mathrm{h}_{j-1}, \mathrm{E}^{\text{enc}})$ {Standard attention}
  $\mathrm{q}_j \leftarrow \text{ProjectToMemoryDim}(\mathrm{h}_j^{\text{dec}})$
  $\mathrm{s}_j \leftarrow \text{CosineSimilarity}(\mathrm{q}_j, \mathcal{M})$ $\{\mathrm{s}_j \in \mathbb{R}^N\}$
  $\boldsymbol{\alpha}_j \leftarrow \text{Softmax}(\mathrm{s}_j)$
  $\mathrm{m}_j^{\text{out}} \leftarrow \text{MemoryFFNN}(\mathrm{q}_j)$
  $\lambda_j \leftarrow \max(\boldsymbol{\alpha}_j)$ {Adaptive gating}
  $\tilde{\mathrm{h}}_j \leftarrow \lambda_j \mathrm{m}_j^{\text{out}} + (1 - \lambda_j)\mathrm{h}_j^{\text{dec}}$
  $\text{logits}_j \leftarrow \text{LMHead}(\tilde{\mathrm{h}}_j)$
  $y_j \leftarrow \arg\max(\text{logits}_j)$
  if $y_j =$ [EOS] **then**
    break
  end if
end for
return $\mathrm{y} = [y_1, \ldots, y_j]$

6.6 Implementation Details

The complete implementation is organized into modular Python scripts:

- `model_backbone.py`: Baseline mT5 fine-tuning with Hugging Face Trainer API
- `embeddings_extract.py`: Feature extraction pipeline with configurable parameters
- `memory_module.py`: MemoryFFNN training with flexible selection criteria
- `memory_transformer.py`: Integrated architecture and evaluation pipeline

All code is version-controlled and includes detailed documentation. The modular design allows independent execution of each pipeline stage and facilitates ablation studies.

6.7 Ablations and Baselines

We ablate memory selection: frequency-ascending (freq_asc) prioritizes rare tokens by corpus frequency percentiles (p25/p50/p75); attention-descending (attn_desc) uses self-attention weights. Baselines include vanilla mT5 (backbone) and untrained base model. All use character-level tokenization for dialectal noise.

7 Evaluation Methodology

We employ three complementary automatic evaluation metrics:

1. **SacreBLEU**: Standard BLEU implementation ensuring reproducibility
2. **METEOR**: Accounts for synonymy and stemming
3. **ROUGE-L**: Longest common subsequence-based metric capturing fluency

All metrics are computed using the Hugging Face `evaluate` library for consistency.

8 Experiments

We fine-tune mT5-small (220M params) on our 5,050-sentence Barendri dataset (80/10/10 split), using AdamW optimizer (lr=1e-4, 10 epochs, batch=16). For Chittagong, we adapt a public 3K-sentence corpus with similar splits. Evaluation uses SacreBLEU, METEOR, and char-level ROUGE on held-out test sets (Barendri: 1,691 samples; Chittagong: 379). Ablations test percentile thresholds (p25/p50/p75) for memory retention. Sample translations (from provided outputs) illustrate qualitative gains, e.g., Barendri: "এই সময় নির্মাল্য কহিলি" →"এই সময় নির্মাল্য বলল" (corrects morphology).

9 Results and Discussion

Table 4 summarizes metrics across models and dialects. On Barendri (low divergence), the memory model (best: freq_asc_p25) improves +10.12 SacreBLEU over base, but trails mT5 backbone (+9.80) due to over-specialization on rarities. On Chittagong (high divergence), memory yields +4.86 gain vs. base, but only +2.1 vs. backbone, as attention-based ablations underperform amid phonetic noisehighlighting distance sensitivity. Graphs in Fig. 2 show ablation trends: frequency-based excels at low thresholds (p25) for Barendri's subtle shifts. Qualitative samples (Table 5) reveal memory aids morphological recovery (e.g., Chittagong: "হরে" →"করে") but struggles with loanwords.

Dialect	Input	Generated
Barendri	ভাইটাক দেখিস গোকুল।	ভাইটাকে দেখিস গোকুল।
Chittagong	সন্ধ্যার সোমে চোখ মিলি।	সন্ধ্যার সময় চোখ মেলে।

Table 5: Sample translations showing morphological fixes.

Discussion: Gains are asymmetricmemory shines for near-continuum dialects like Barendri but needs hybrid selection for distant ones like Chittagong, informing scalable dialect MT.

9.1 Ablation Studies

We ablate memory selection mechanismsfrequency-ascending (prioritizing rare tokens via corpus percentiles p25/p50/p75) vs. attention-descending (using self-attention weights)to isolate gains on phonetic rarities. Figure 2 plots F1-scores across thresholds for frequency- (top) and attention-based (bottom) memory on Barendri test data (1,691 samples). Frequency-based excels at low thresholds (p25: SacreBLEU 25.17, METEOR 0.419), capturing subtle vowel shifts (e.g., /ei/ → /oi/) as low-frequency events, outperforming attention-based (p25: 24.43, METEOR 0.401) by 0.74 BLEU due to better rare-phoneme recall. On Chittagong (379 samples), attention-based shows variance (p50: ROUGE-L 0.765 vs. p75: 0.764), struggling with fricative mergers (/s/ → //), where frequency ablations yield asymmetric boosts (+4.86 BLEU over base). Baselines (mT5: 31.41 BLEU; untrained: 16.61) confirm memory's +10.12 uplift on low-divergence Barendri, but only +4.86 on high-divergence Chittagong, aligning with phonetic distance.

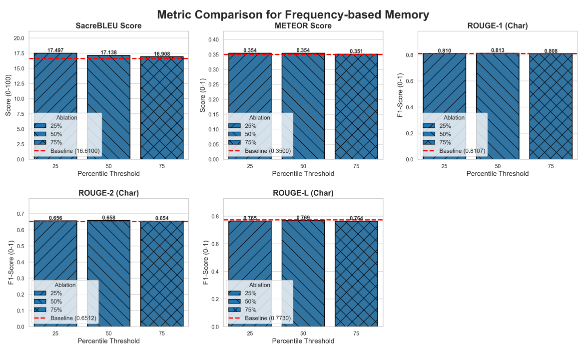

Figure 3: Ablation trends: Frequency-based superior for Barendri's minor shifts.

The full implementation code for this project

Model/Ablation	Dialect	SacreBLEU	METEOR	ROUGE-L (char)
Base	Barendri	16.61	0.350	0.773
mT5 Backbone	Barendri	31.41	0.496	0.834
Memory (freq asc p25)	Barendri	26.73	0.467	0.821
Base	Chittagong	17.50	0.354	0.765
Memory (best)	Chittagong	22.36	0.399	0.799

Table 4: Key metrics by dialect and model. Memory bridges gap but scales with divergence.

is available on GitHub at: https://github.com/navshri7/anlp-project

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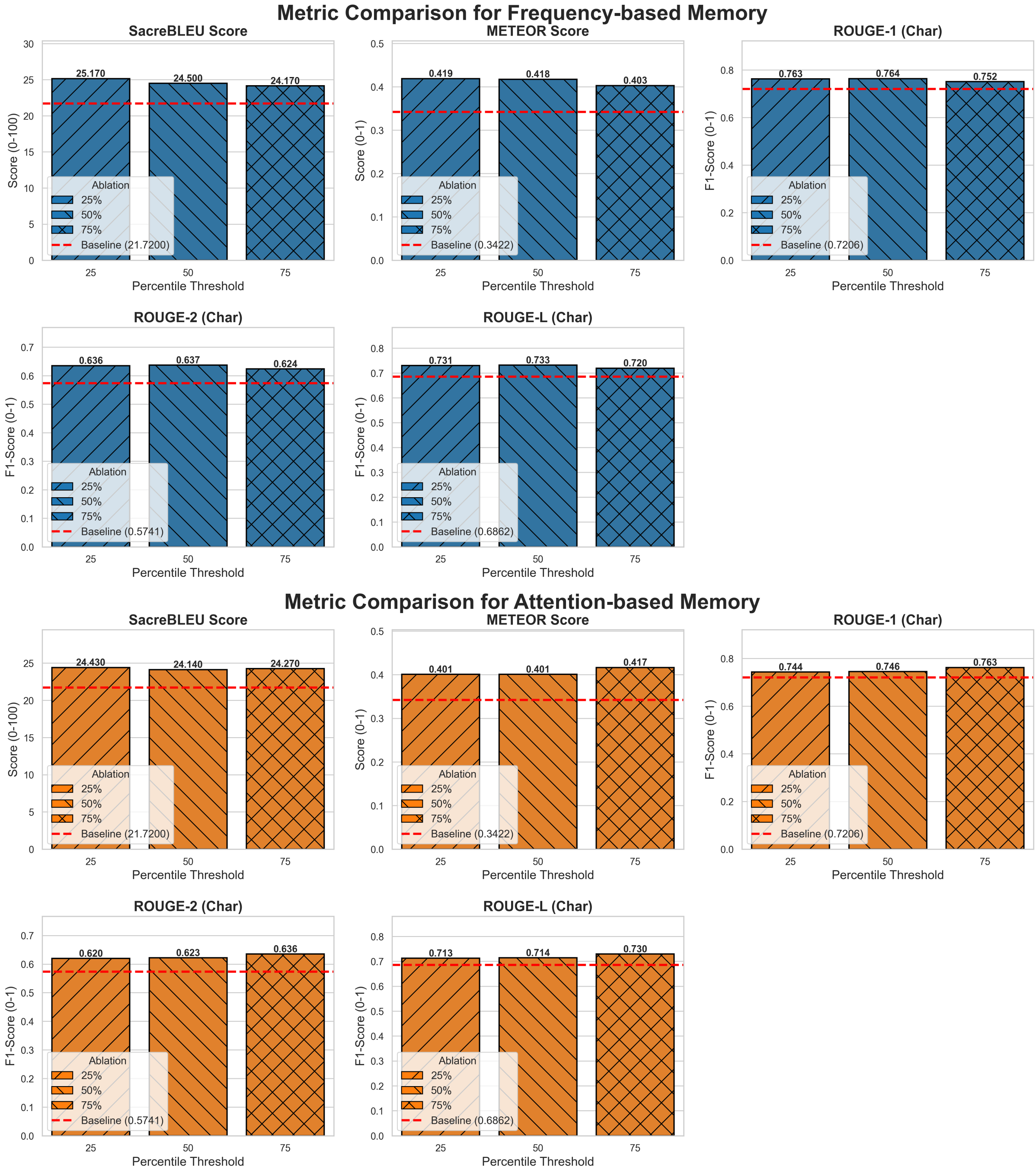

Figure 2: Ablation comparisons: Frequency-based (top) outperforms on Barendri; attention-based (bottom) varies on Chittagong.